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Chua et al.

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(54) **SOLE STRUCTURE FOR ARTICLE OF FOOTWEAR**

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(65) **Prior Publication Data**

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Related U.S. Application Data

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(Continued)

(57) **ABSTRACT**

A sole structure for an article of footwear includes a first annular group of traction elements arranged along a first annular zone and a second annular group of traction elements arranged along a second annular zone concentric with the first annular group. The first and second annular groups of traction elements include a plurality of directional traction elements arranged in a first rotational direction about a common rotation zone. Optionally, the first annular group of traction elements may include an omnidirectional traction element arranged at a location associated with a relatively low degree of alignment between radii of rotation corresponding to different torsional movements of the sole structure during use. The directional traction elements may include unidirectional traction elements or bidirectional traction elements at locations associated with moderate to high degrees of alignment between radii of rotation corresponding to the different torsional movements.

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A43C 15/16 (2006.01)

A43B 13/22 (2006.01)

(52) **U.S. Cl.**

CPC **A43C 15/161** (2013.01); **A43B 13/223** (2013.01); **A43C 15/162** (2013.01)

(58) **Field of Classification Search**

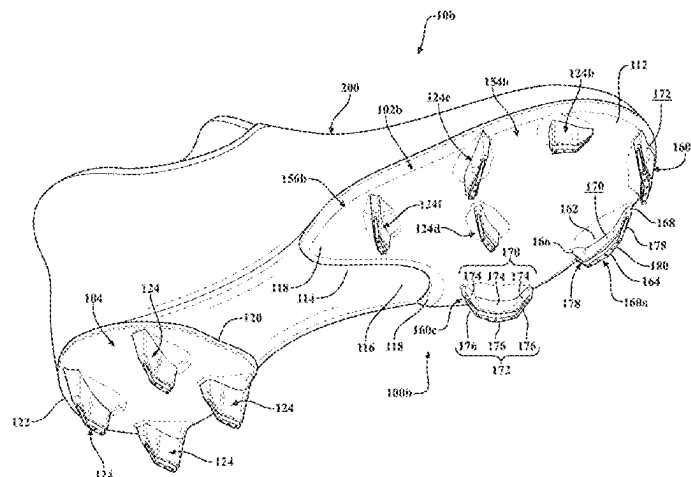
CPC **A43C 15/161–168**; **A43B 13/223**
See application file for complete search history.

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20 Claims, 16 Drawing Sheets



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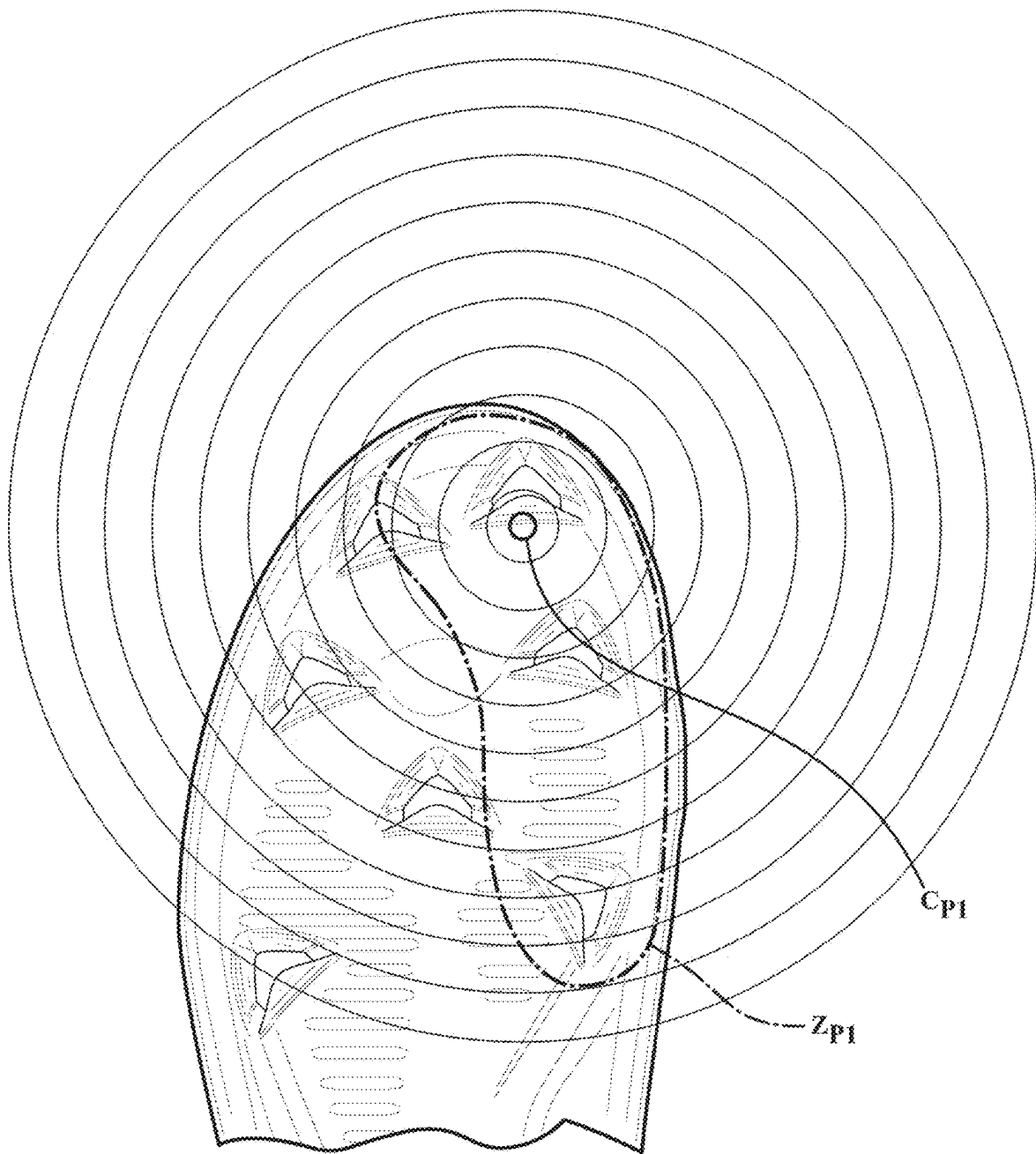
FIG. 1A

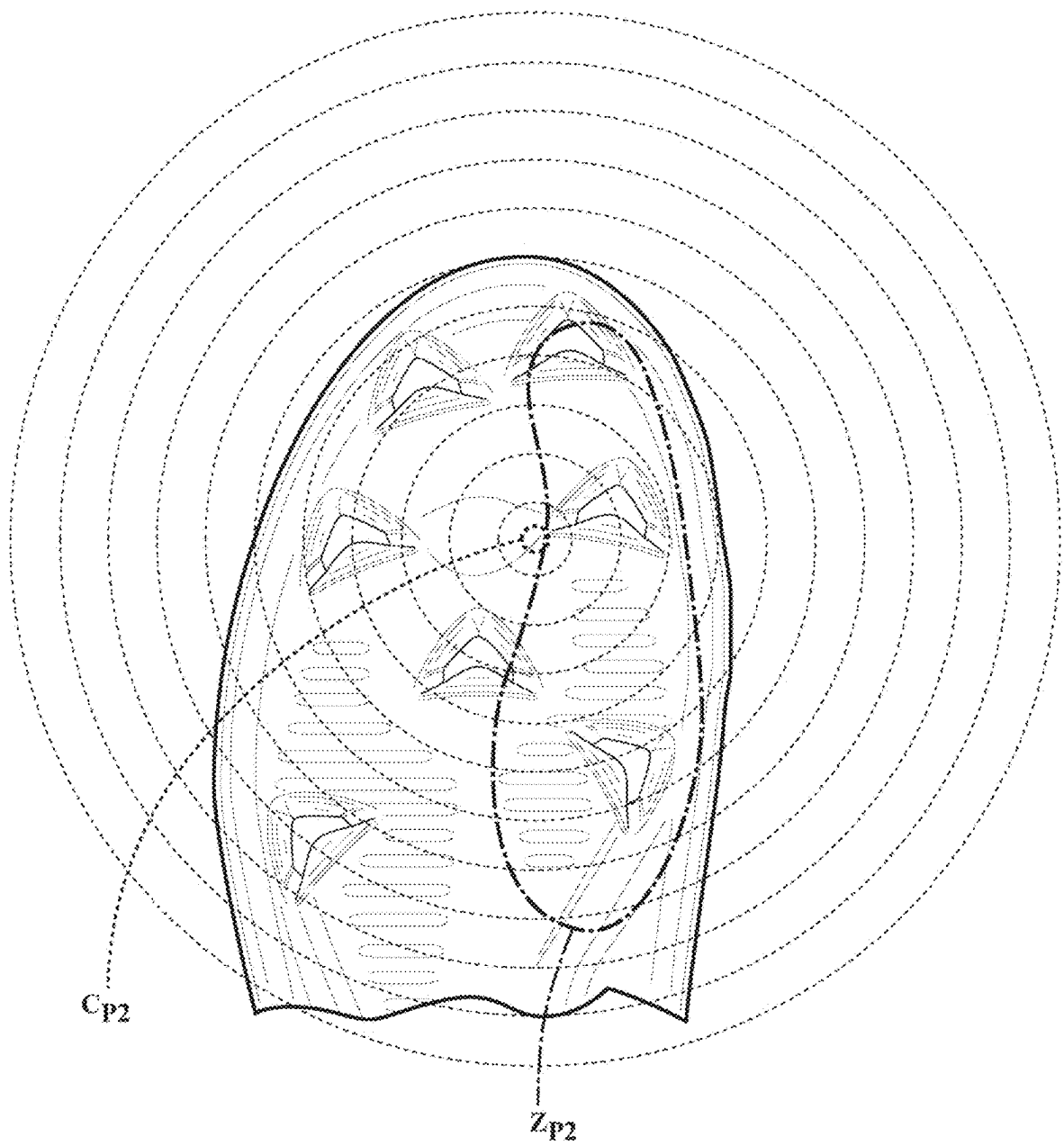
FIG. 1B

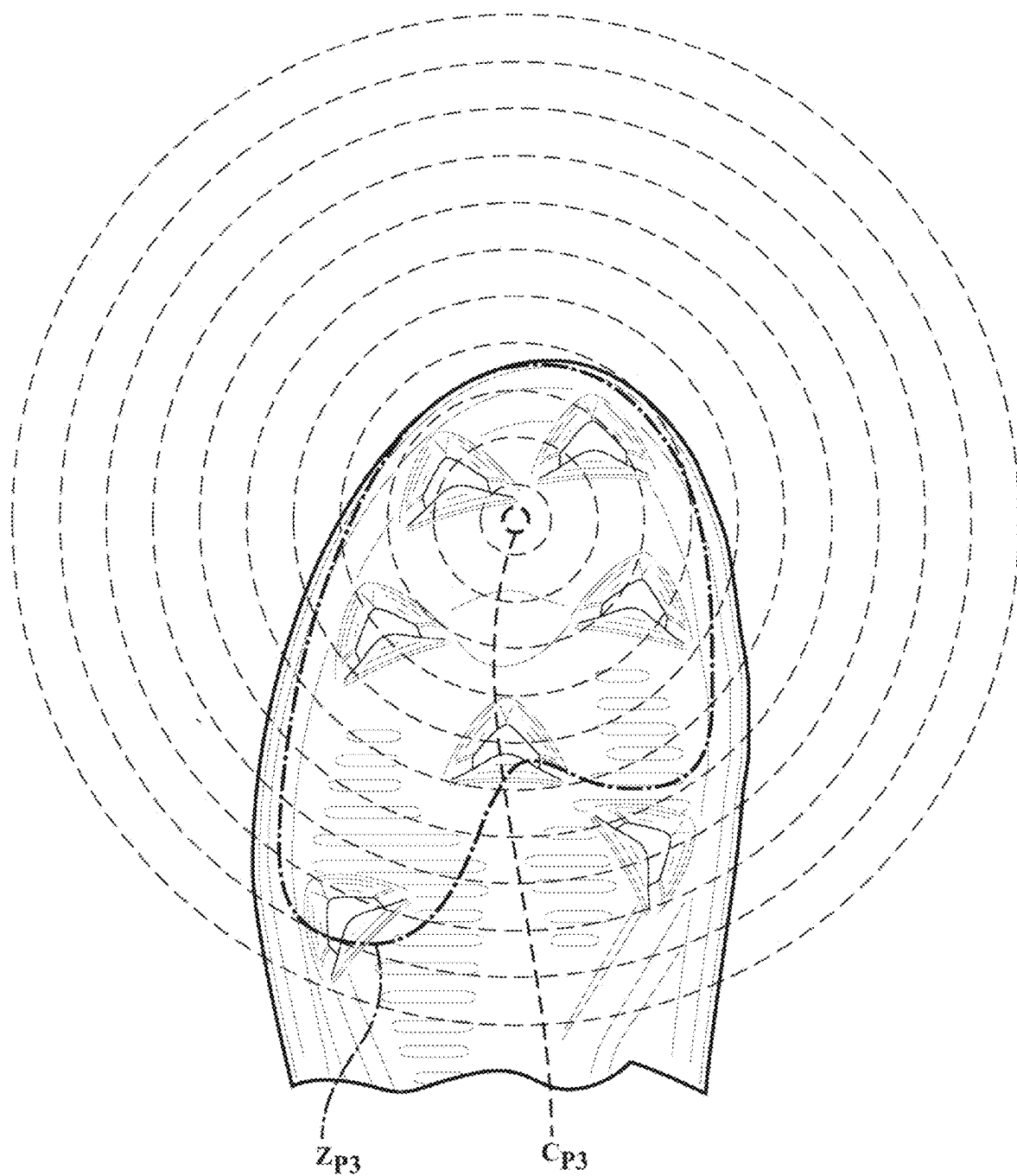
FIG. 1C

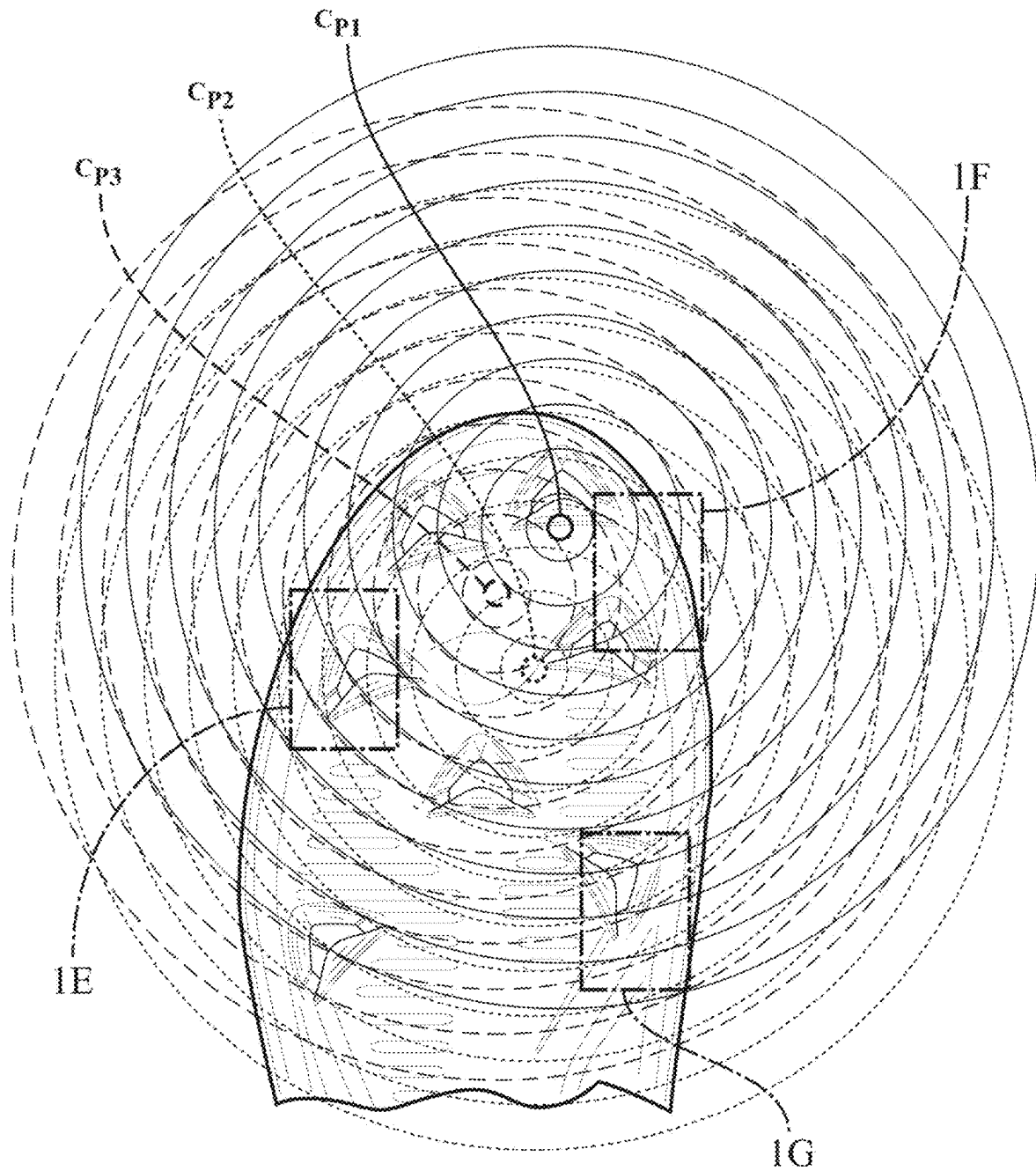
FIG. 1D

FIG. 1E

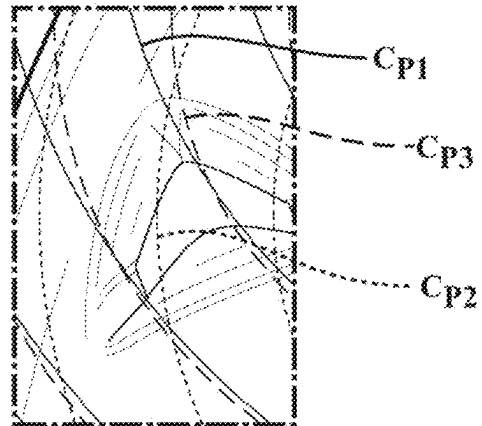


FIG. 1F

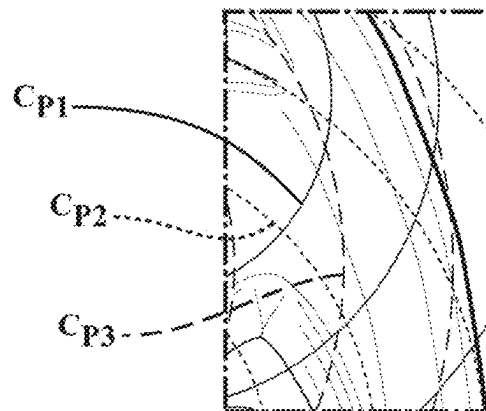
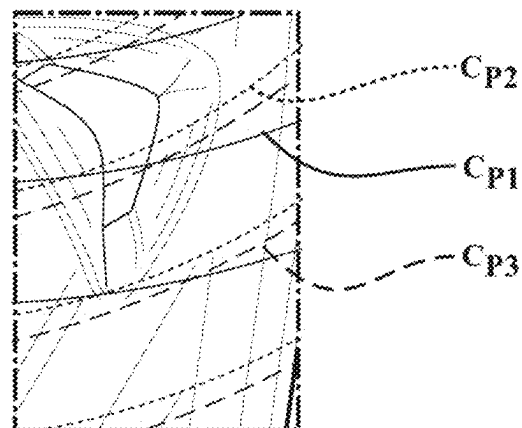
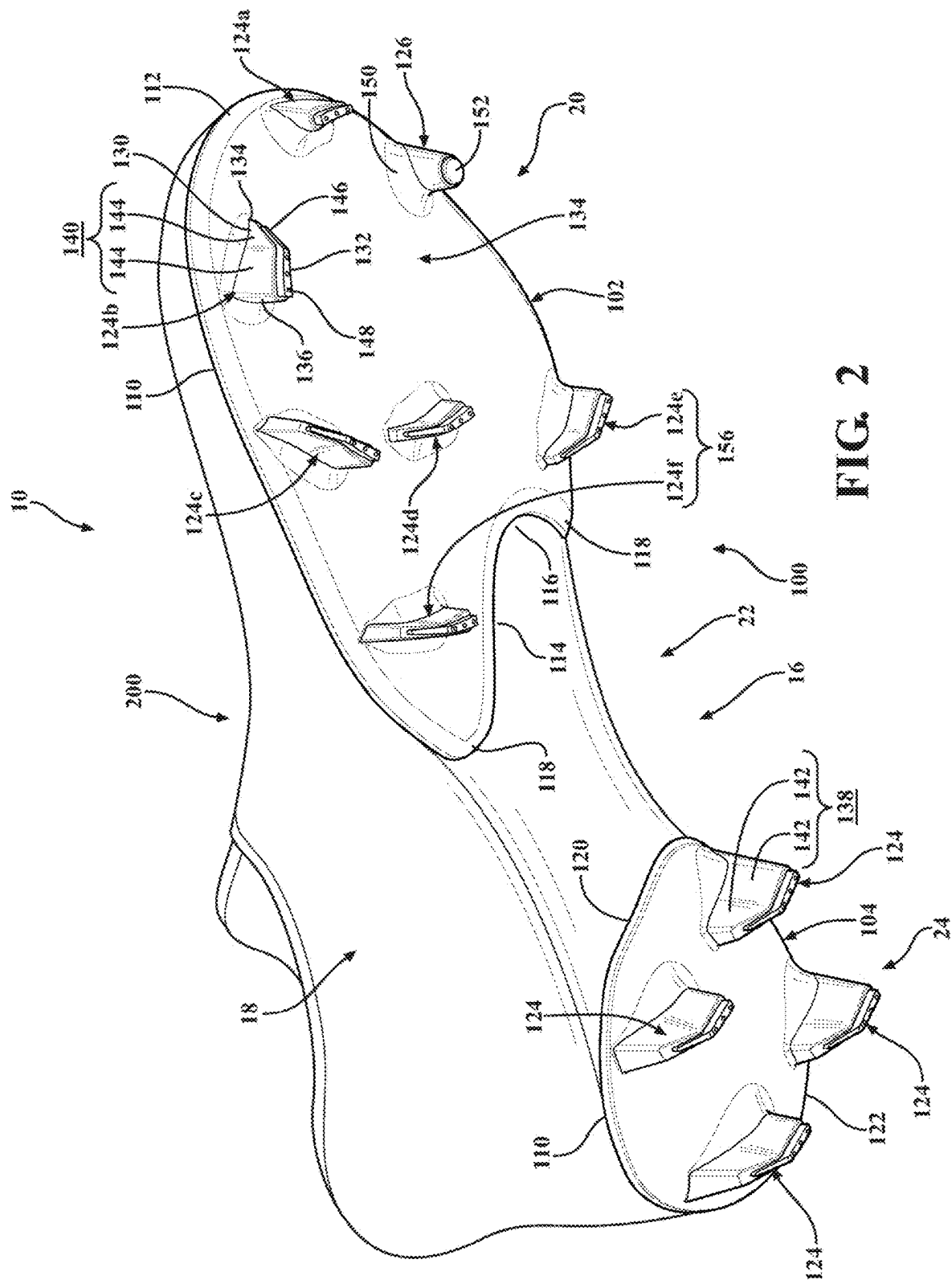
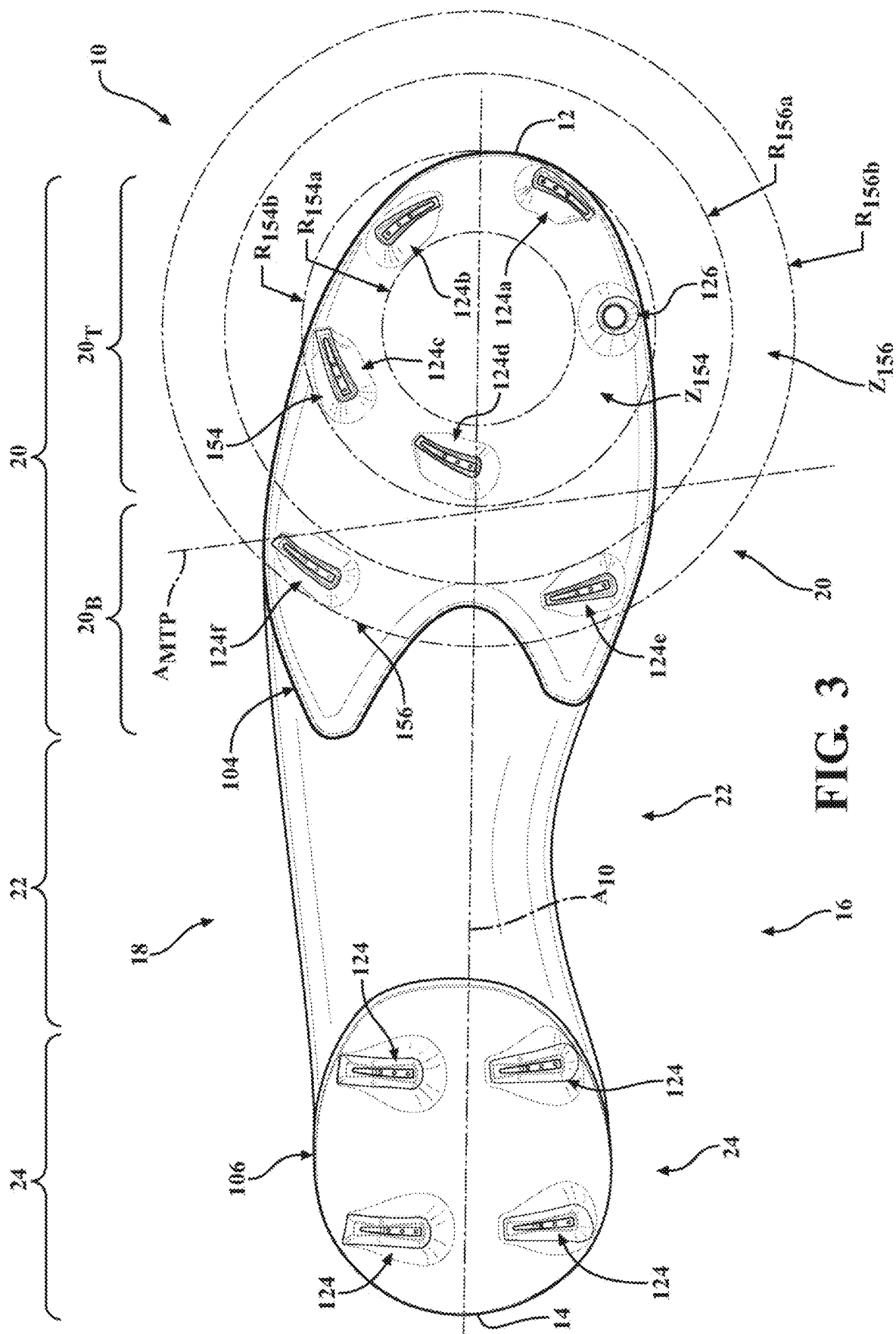


FIG. 1G







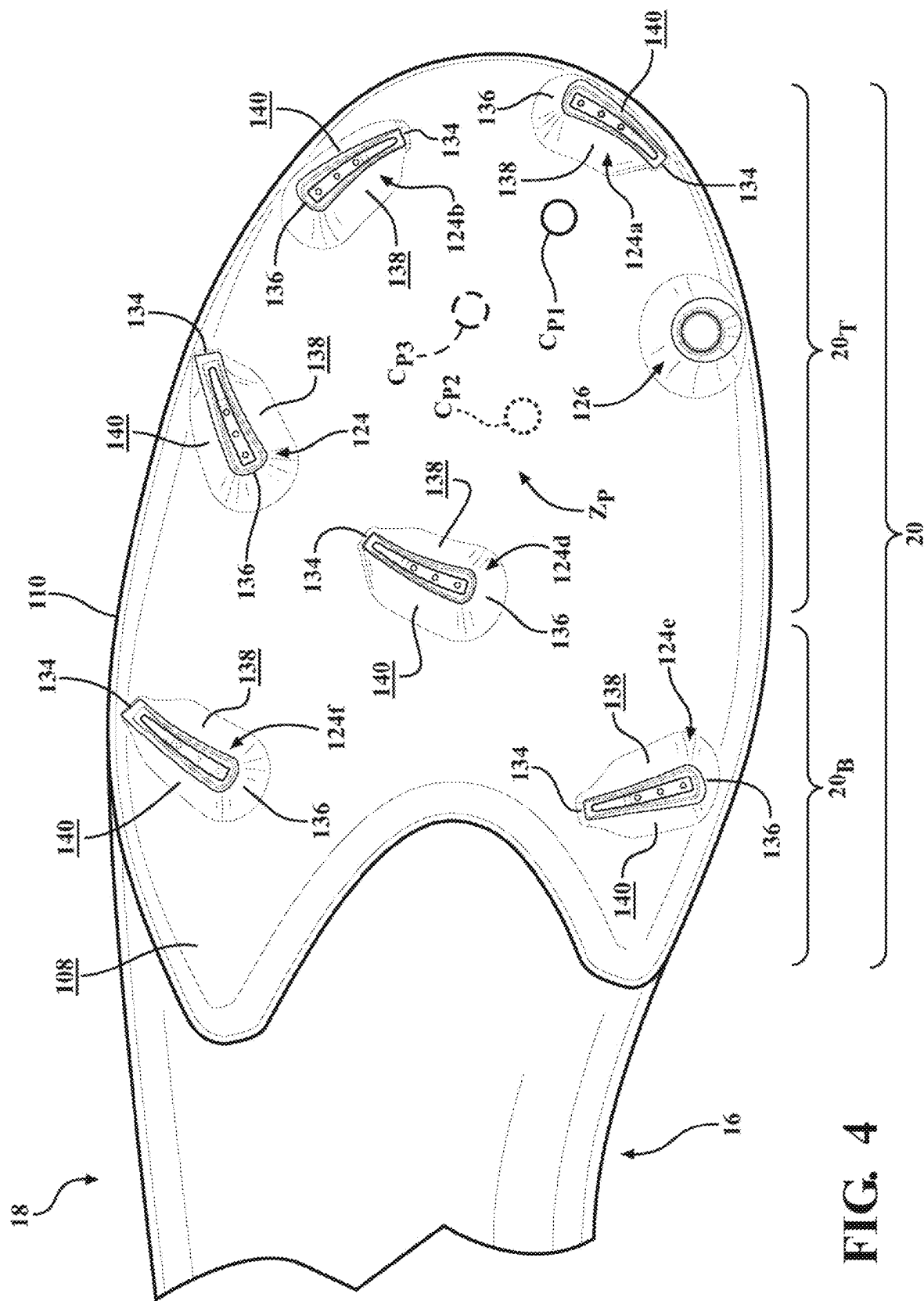


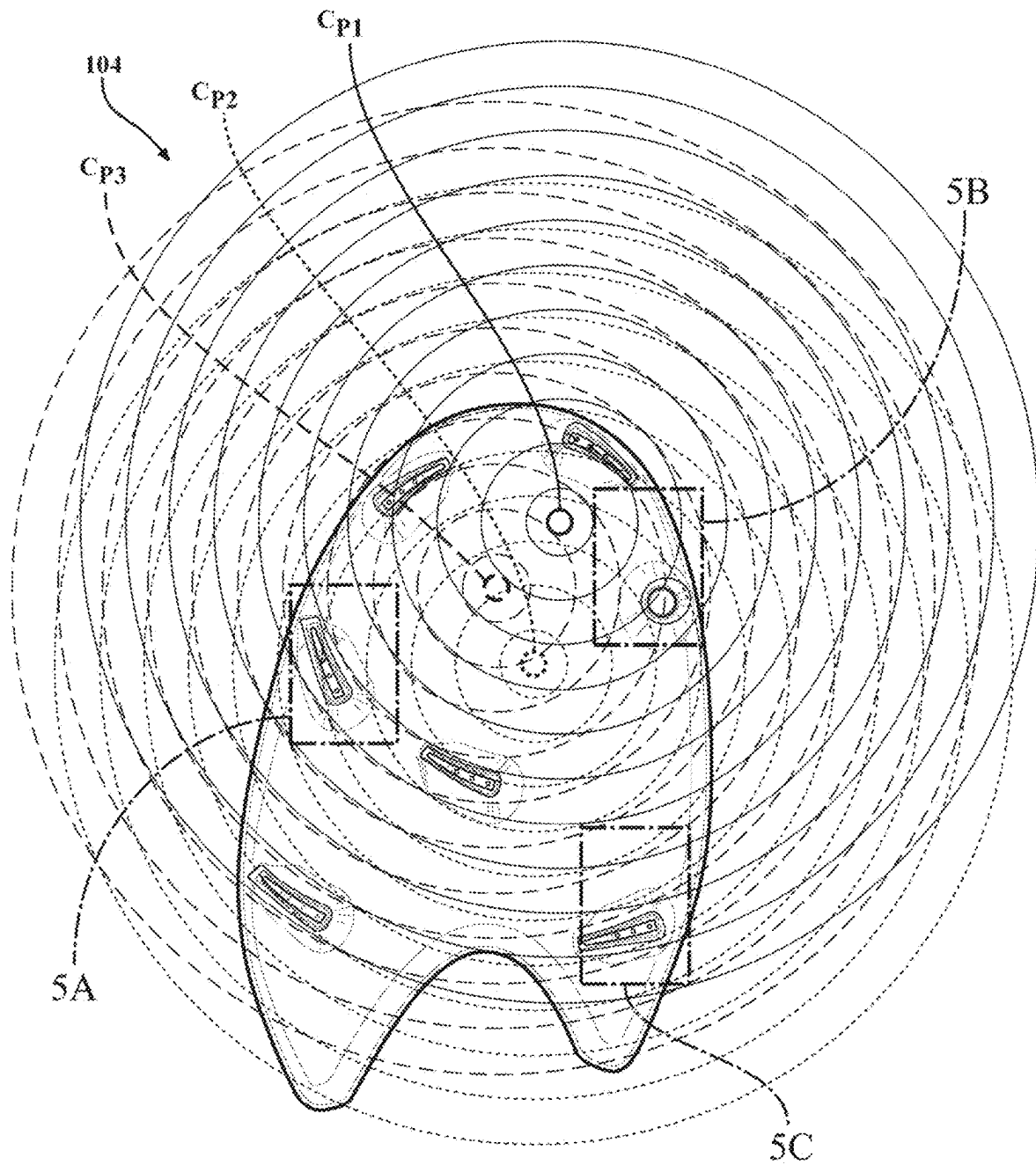
FIG. 5

FIG. 5A

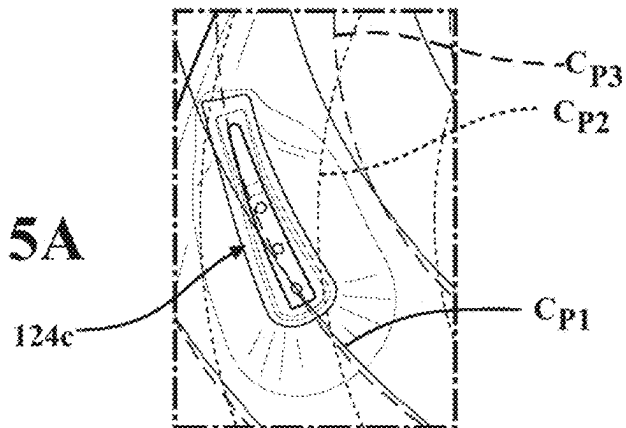


FIG. 5B

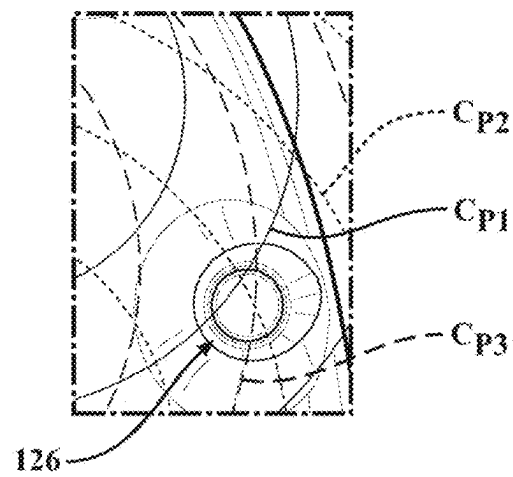
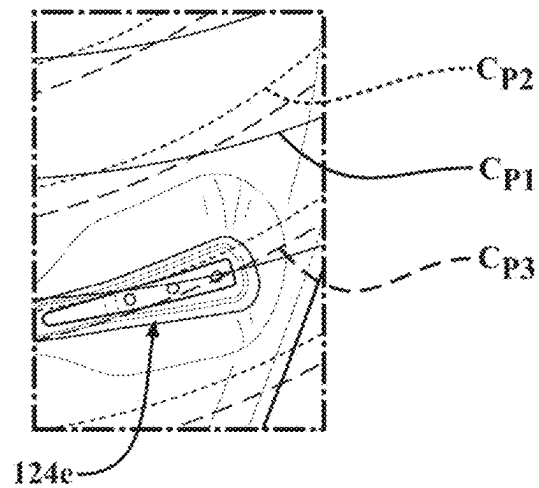
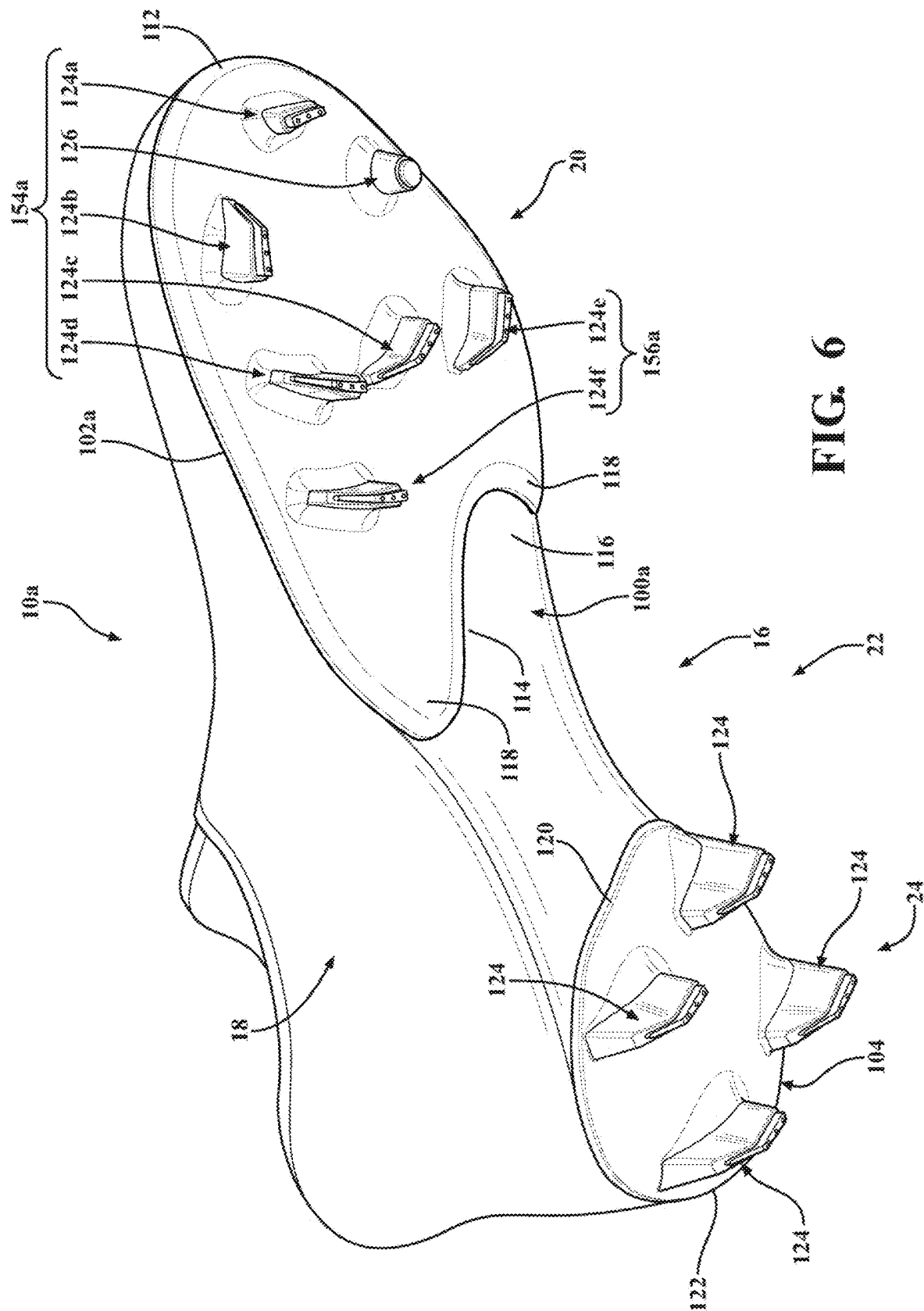


FIG. 5C





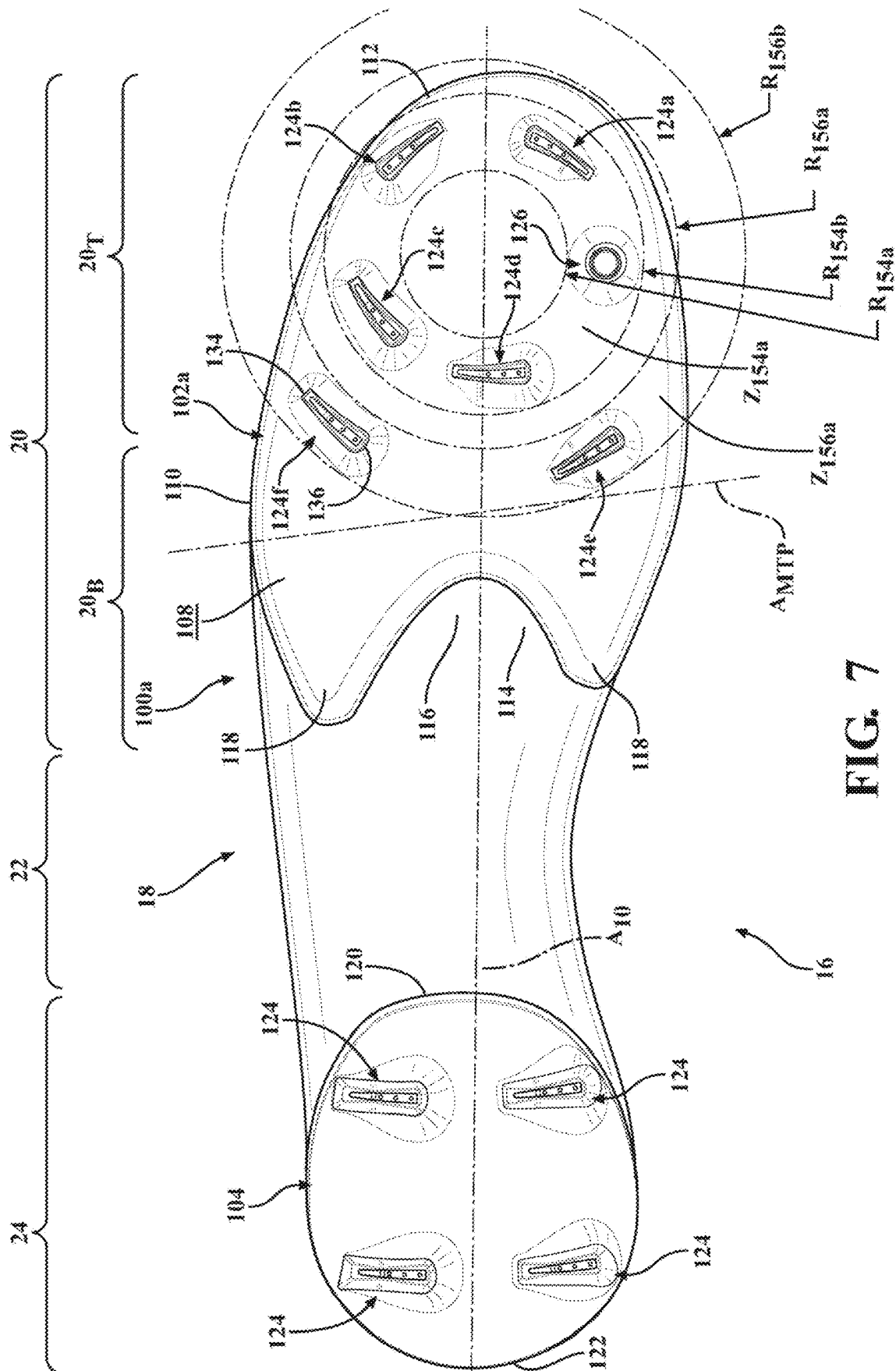
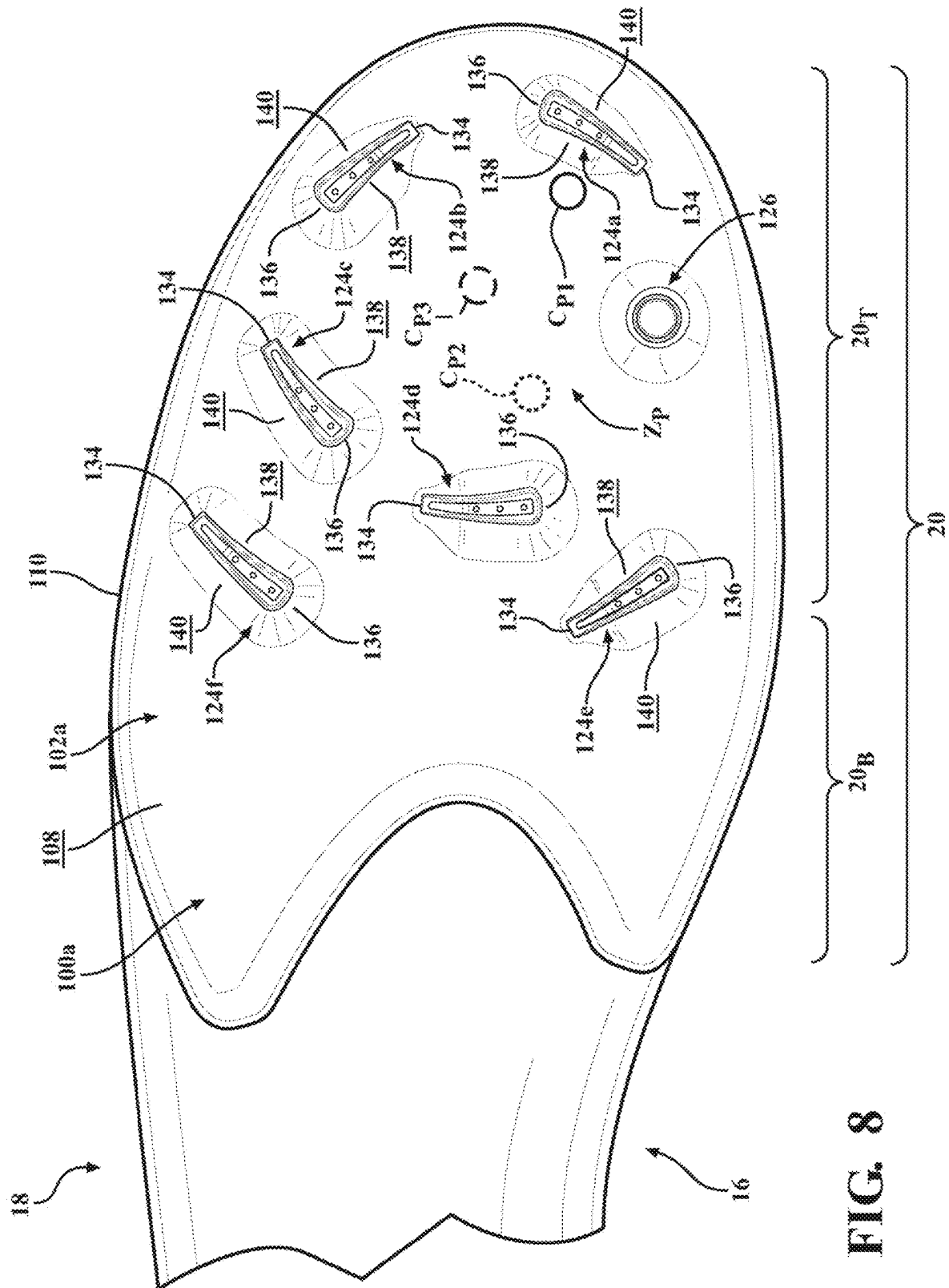


FIG. 7



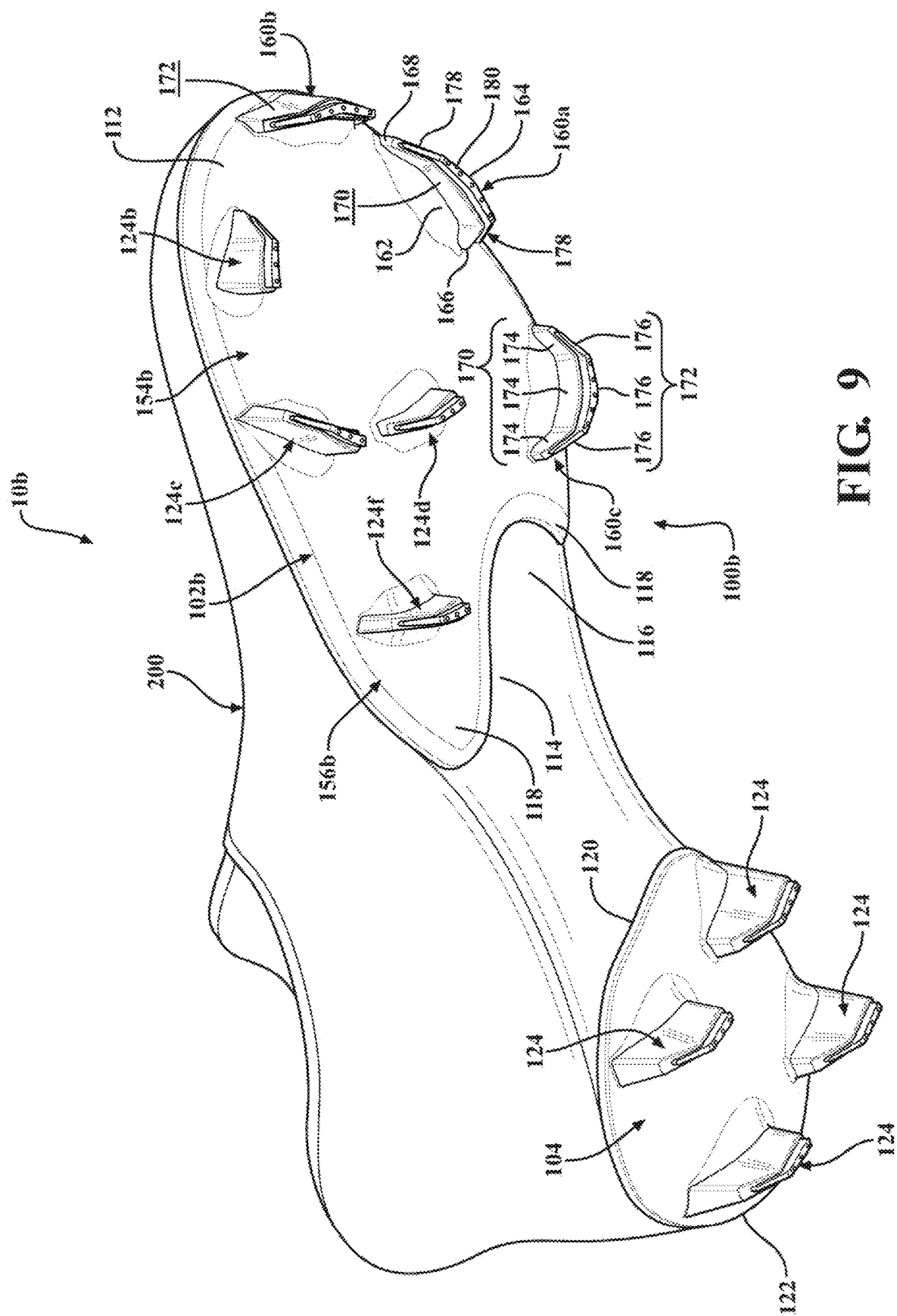
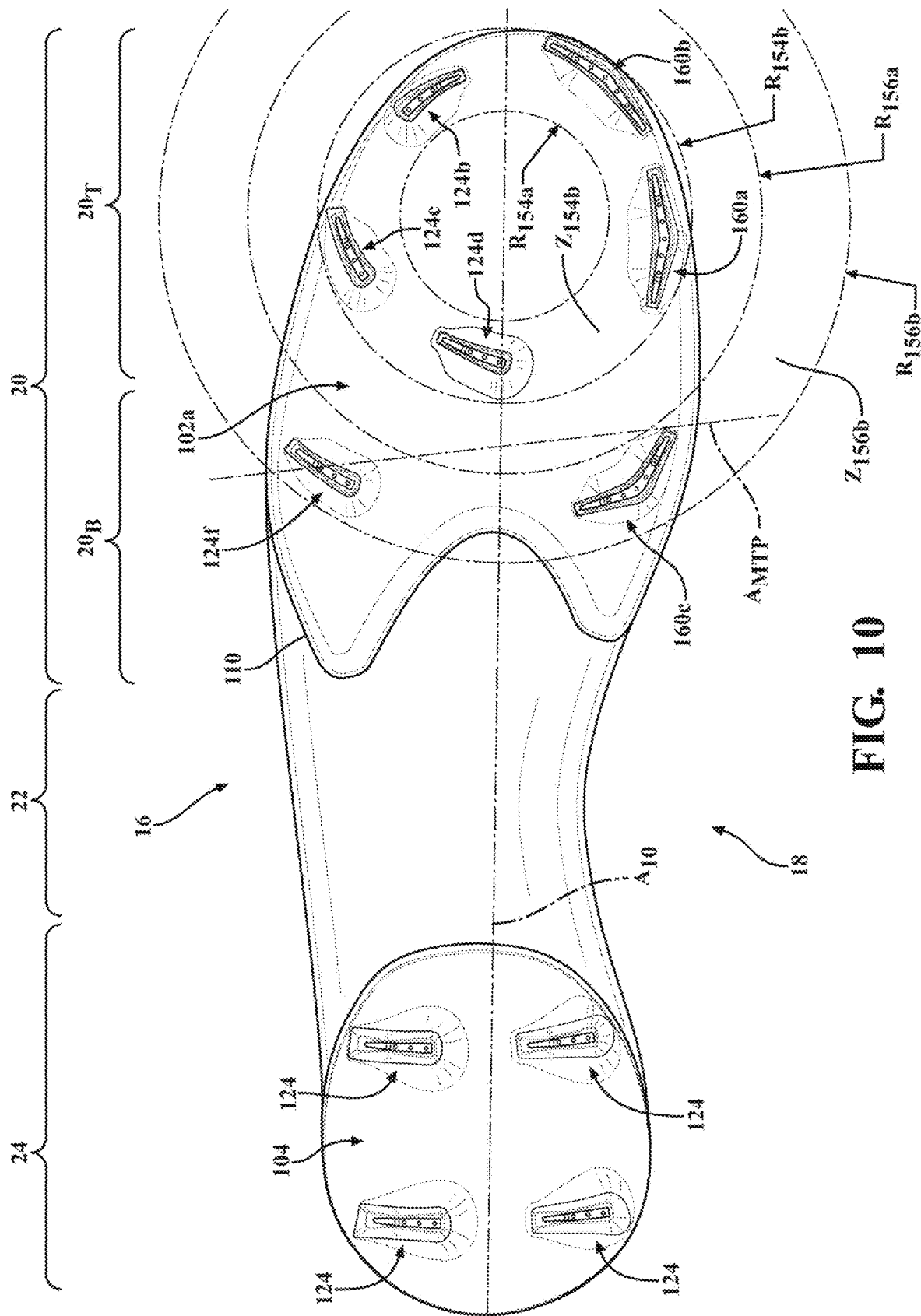


FIG. 9



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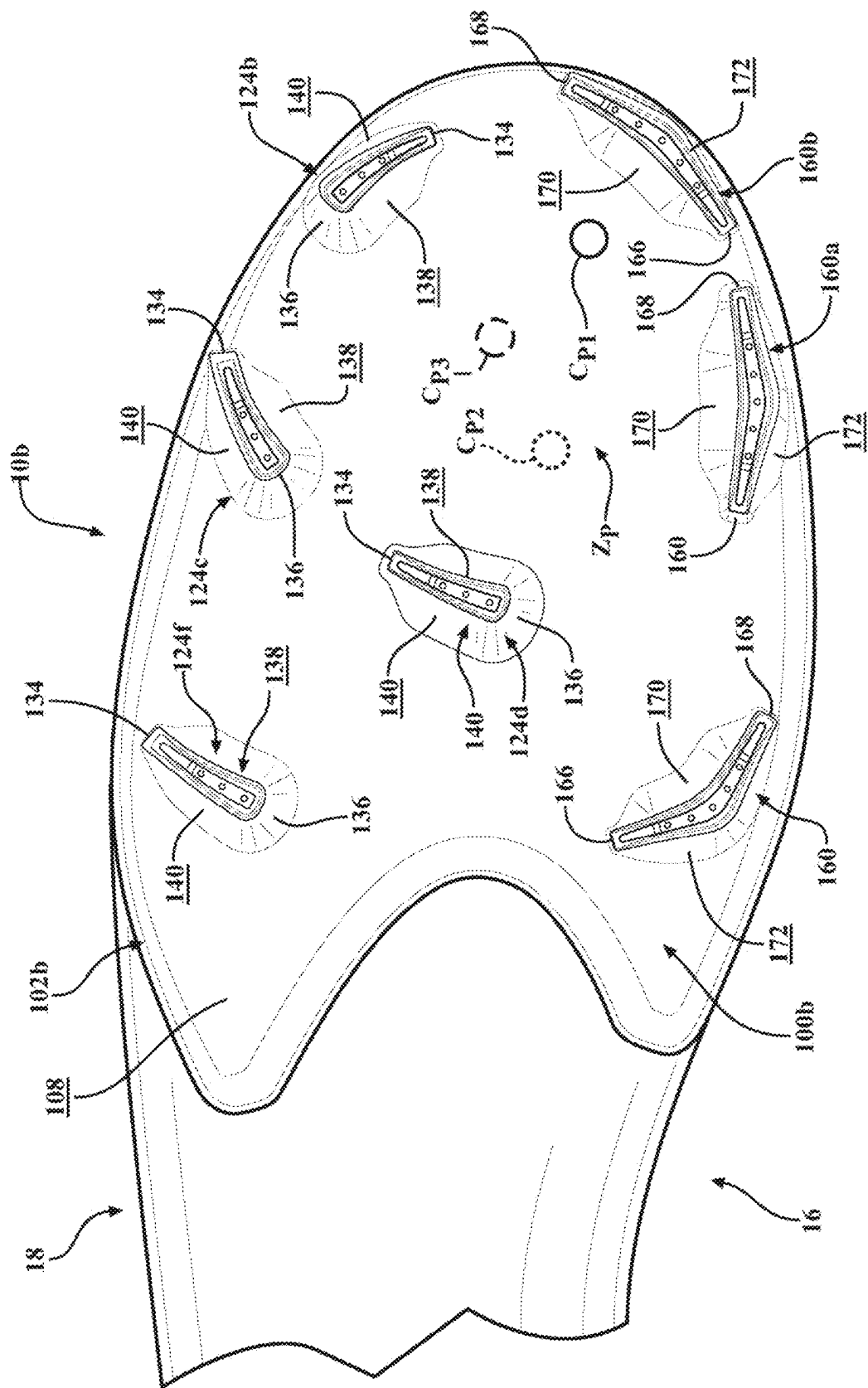


FIG. 11

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SOLE STRUCTURE FOR ARTICLE OF FOOTWEAR

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/470,790, filed on Sep. 9, 2021, which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Application No. 63/077,208, filed on Sep. 11, 2020. The disclosures of these prior applications are considered part of the disclosure of this application and are hereby incorporated by reference in their entirety.

FIELD

The present disclosure relates generally to an article of footwear, and more particularly to a sole structure for an article of footwear.

BACKGROUND

This section provides background information related to the present disclosure and is not necessarily prior art.

Articles of footwear conventionally include an upper and a sole structure. The upper may be formed from any suitable material(s) to receive, secure, and support a foot on the sole structure. The upper may cooperate with laces, straps, or other fasteners to adjust the fit of the upper around the foot. A bottom portion of the upper, proximate to a bottom surface of the foot, attaches to the sole structure.

Sole structures generally include a layered arrangement extending between a ground surface and the upper. For example, a sole structure may include a midsole and an outsole. The midsole is generally disposed between the outsole and the upper and provides cushioning for the foot. The midsole may include a pressurized fluid-filled chamber that compresses resiliently under an applied load to cushion the foot by attenuating ground-reaction forces. The outsole provides abrasion-resistance and traction with the ground surface and may be formed from rubber or other materials that impart durability and wear-resistance, as well as enhancing traction with the ground surface.

While known outsoles have proven acceptable for their intended purposes, a continuous need for improvement in the relevant art remains. For example, a need exists for an outsole that provides improved traction with the ground surface when forces having varying magnitude and direction are applied from the midsole or the upper to the outsole. A need also exists for an article of footwear having improved overall comfort and fit while providing such improved traction.

DRAWINGS

The drawings described herein are for illustrative purposes only of selected configurations and not all possible implementations, and are not intended to limit the scope of the present disclosure.

FIG. 1A is an environmental view illustrating a contact zone and centroid associated with an article of footwear and resulting from a first movement;

FIG. 1B is an environmental view illustrating a contact zone and centroid associated with an article of footwear and resulting from a second movement;

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FIG. 1C is an environmental view illustrating a contact zone and centroid associated with an article of footwear and resulting from a third movement;

FIG. 1D is an environmental view showing the centroids associated with the first, second, and third movements overlaid onto an article of footwear;

FIGS. 1E-1G are enlarged environmental views showing radii associated with the first, second, and third centroids and respectively taken at Areas 1E-1G of FIG. 1D;

FIG. 2 is a bottom perspective view of an example of an article of footwear according to the present disclosure;

FIG. 3 is a bottom plan view of the article of footwear of FIG. 2;

FIG. 4 is an enlarged fragmentary view of the article of footwear of FIG. 2, showing a forefoot region of the article of footwear;

FIG. 5 is an environmental view of a forefoot plate of the article of footwear of FIG. 2 overlaid with the centroids and radii of rotation associated with the first, second, and third movements of FIG. 1D;

FIGS. 5A-5C are enlarged environmental views respectively taken at Areas 5A-5C of FIG. 5 and showing the radii of rotation associated with the first, second, and third movements;

FIG. 6 is a bottom perspective view of an example of an article of footwear according to the present disclosure;

FIG. 7 is a bottom plan view of the article of footwear of FIG. 6;

FIG. 8 is an enlarged fragmentary view of the article of footwear of FIG. 6, showing a forefoot region of the article of footwear;

FIG. 9 is a bottom perspective view of an example of an article of footwear according to the present disclosure;

FIG. 10 is a bottom plan view of the article of footwear of FIG. 9; and

FIG. 11 is an enlarged fragmentary view of the article of footwear of FIG. 9, showing a forefoot region of the article of footwear;

Corresponding reference numerals indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

Example configurations will now be described more fully with reference to the accompanying drawings. Example configurations are provided so that this disclosure will be thorough, and will fully convey the scope of the disclosure to those of ordinary skill in the art. Specific details are set forth such as examples of specific components, devices, and methods, to provide a thorough understanding of configurations of the present disclosure. It will be apparent to those of ordinary skill in the art that specific details need not be employed, that example configurations may be embodied in many different forms, and that the specific details and the example configurations should not be construed to limit the scope of the disclosure.

The terminology used herein is for the purpose of describing particular exemplary configurations only and is not intended to be limiting. As used herein, the singular articles “a,” “an,” and “the” may be intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “comprising,” “including,” and “having,” are inclusive and therefore specify the presence of features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. The method steps, processes, and operations

described herein are not to be construed as necessarily requiring their performance in the particular order discussed or illustrated, unless specifically identified as an order of performance. Additional or alternative steps may be employed.

When an element or layer is referred to as being “on,” “engaged to,” “connected to,” “attached to,” or “coupled to” another element or layer, it may be directly on, engaged, connected, attached, or coupled to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly engaged to,” “directly connected to,” “directly attached to,” or “directly coupled to” another element or layer, there may be no intervening elements or layers present. Other words used to describe the relationship between elements should be interpreted in a like fashion (e.g., “between” versus “directly between,” “adjacent” versus “directly adjacent,” etc.). As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

The terms first, second, third, etc. may be used herein to describe various elements, components, regions, layers and/or sections. These elements, components, regions, layers and/or sections should not be limited by these terms. These terms may be only used to distinguish one element, component, region, layer or section from another region, layer or section. Terms such as “first,” “second,” and other numerical terms do not imply a sequence or order unless clearly indicated by the context. Thus, a first element, component, region, layer or section discussed below could be termed a second element, component, region, layer or section without departing from the teachings of the example configurations.

The details of one or more implementations of the disclosure are set forth in the accompanying drawings and the description below. Other aspects, features, and advantages will be apparent from the description, the drawings, and the claims.

In one aspect of the disclosure, a structure for an article of footwear includes a first plurality of traction elements having a first series of directional traction elements arranged within a first annular zone in a first rotational direction around a pivot zone. The sole structure further includes a second plurality of traction elements including a second series of directional traction elements arranged within a second annular zone concentric with and larger than the first annular zone. The second plurality of traction elements are arranged in the first rotational direction around the pivot zone.

Aspects of the disclosure may include one or more of the following optional features. In some examples, each of the directional traction elements is elongate and includes a length extending from a first end to a second end along the first rotational direction. In some implementations, each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

In some configurations, each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction. Optionally, the inner surface converges with the outer surface along the first rotational direction.

In some implementations, the sole structure includes a third plurality of directional traction elements disposed in a heel region, each of the directional traction elements of the third plurality of directional traction elements oriented in the first rotational direction. In some examples, the first plurality

of traction elements and the second plurality of traction elements are disposed in a forefoot region of the sole structure.

In some configurations, the first plurality of traction elements further includes an omnidirectional traction element arranged within the first annular zone. Here, the omnidirectional traction element is disposed on a medial side of the sole structure and at least one of the directional traction elements of the first series is disposed on a lateral side of the sole structure. In some examples, at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

Another aspect of the disclosure provides a sole structure for an article of footwear including a first annular group of traction elements and a second annular group of traction elements. The first annular group of traction elements is arranged in series along a first annular zone in a forefoot region. The first annular group includes a first directional traction element on a lateral side of the sole structure and a second directional traction element on a medial side of the sole structure. The second annular group of traction elements is arranged in series along a second annular zone concentric with the first annular zone. The second annular group of traction elements includes a third directional traction element on the lateral side of the sole structure and a fourth directional traction element on the medial side of the sole structure.

Aspects of the disclosure may include one or more of the following optional features. In some examples, each of the directional traction elements is elongate and includes a length extending from a first end to a second end along a first rotational direction around a pivot zone of the sole structure. Optionally, each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

In some implementations, each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction. In some examples, the inner surface converges with the outer surface along the first rotational direction. Optionally, the sole structure includes a third group of directional traction elements disposed in a heel region, each of the directional traction elements of the third group of directional traction elements oriented in the first rotational direction. In some configurations, the first annular group and the second annular group are disposed in the forefoot region of the sole structure.

In some examples, the first annular group of traction elements further includes an omnidirectional traction element arranged along the first annular zone. Here, the omnidirectional traction element may be disposed on a medial side of the sole structure. In some implementations, at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

FIGS. 1A-1F illustrate examples of reactionary forces and motions corresponding to common athletic movements associated with an article of footwear. As shown, these forces and motions are illustrated with respect to an article of footwear including a conventional pattern of traction elements configured for translational (i.e., lateral, longitudinal) traction with a ground surface. In FIG. 1A, a first contact zone Z_{P1} associated with a 45° outside cut (i.e., forward and lateral direction) is shown. Here, the first contact zone Z_{P1} indicates a pressure area along the anterior-medial side of the article of footwear, with a higher degree

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of contact at the anterior end associated with the 45° movement. A centroid C_{P1} associated with the first contact zone Z_{P1} is located adjacent to an anterior end of the article of footwear. FIG. 1B illustrates a second contact zone Z_{P2} and second centroid C_{P2} corresponding to a second movement associated with a 180° outside cut (i.e., lateral direction). As shown, the second contact zone Z_{P2} and second centroid C_{P2} are shifted away from the anterior end and towards the lateral side relative to the first contact zone Z_{P1} and first centroid C_{P1} . FIG. 1C illustrates a third contact zone Z_{P3} and third centroid C_{P3} corresponding to a third movement associated with a 90° inside cut (i.e., longitudinal direction), such as an acceleration, deceleration, or planting kick. Here, the third contact zone Z_{P3} includes a full width of the sole structure such that the third centroid C_{P3} is closer to the lateral side than the first and second centroids C_{P1} , C_{P2} . For clarity, FIGS. 1A-1C each show a plurality of radii of rotation associated with the respective centroids C_{P1} , C_{P2} , C_{P3} , which will be discussed in greater detail with respect to FIGS. 1D-1G. As discussed throughout the application, the centroids C_{P1} , C_{P2} , C_{P3} may collectively define a rotational or pivot zone Z_P , which is an area of the sole structure encompassing all three of the centroids C_{P1} , C_{P2} , C_{P3} about which the sole structure may pivot during any one of the three movements.

FIGS. 1D-1G illustrate the relationships between the rotational motions corresponding to each of the first, second, and third centroids C_{P1} , C_{P2} , C_{P3} associated with the first, second, and third movements. In FIG. 1D, the rotational radii corresponding to each of the centroids C_{P1} , C_{P2} , C_{P3} are all overlaid upon the same example sole structure. As shown, the rotational radii are offset relative to one another such that the rotational radii have different degrees of alignment at different areas of the sole structure. For example, at Area 1E (FIG. 1E), the rotational radii associated with the first centroid C_{P1} (solid line) and the third centroid C_{P3} (dashed line) have a relatively high degree of alignment (e.g., tangentially) to each other while the rotational radius associated with the second centroid C_{P2} (solid line) extends transversely to the rotational radii associated with the second centroid C_{P2} and the third centroid C_{P3} . Accordingly, at Area 1E, the sole structure moves in a similar rotational direction during a 45° outside cut and a 90° inside cut, but moves along a different rotational path during a 180° outside cut. In another example, at Area 1F (FIG. 1F), the rotational radii associated with the first, second, and third centroids C_{P1} , C_{P2} , C_{P3} have a relatively low degree of alignment with each other. Thus, the sole structure moves in different rotational directions at Area 1F during a 45° outside cut, a 180° outside cut, and a 90° inside cut. At Area 1G (FIG. 1G) the rotational radii associated with the first, second, and third centroids C_{P1} , C_{P2} , C_{P3} have a relatively high degree of alignment with each other, indicating that the sole structure moves along a common rotational path at Area 1G during a 45° outside cut, a 180° outside cut, and a 90° inside cut.

With continued reference to FIGS. 1D-1G, the different degrees of alignment between the rotational radii of the first, second, and third centroids C_{P1} , C_{P2} , C_{P3} may affect a torsional force associated with the article of footwear based on the shape, location, and orientation of traction elements on the article of footwear. For instance, the example of the article of footwear shown in FIGS. 1D-1G includes chevron-shaped traction elements arranged in a random pattern with broad faces of the traction elements being transverse to the rotational radii. Thus, in areas of the sole structure having a relatively high degree of alignment (e.g., Areas 1E, 1G),

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conventional configurations and shapes of traction elements may impart a higher torsional force during the associated movements (e.g., cuts, kicks, etc.) as the traction elements engage the ground surface along the direction of the rotational radii.

As discussed in greater detail below, the examples of the articles of footwear **10**, **10a**, **10b** according to the present disclosure are configured to tune torsional forces associated with the articles of footwear **10**, **10a**, **10b** by providing an annular series of traction elements that are aligned with one another based on an optimized alignment with the rotational radii. Here, combinations of directional traction elements and omnidirectional traction elements are incorporated based on the relationship between the rotational radii. For instance, directional traction elements are provided in areas of the sole structure associated with relatively high degrees of alignment between rotational radii, while omnidirectional traction elements are provided in areas of the sole structure associated with relatively low degrees of alignment between rotational radii.

Referring to FIG. 2, an article of footwear **10** includes a sole structure **100** and an upper **200** attached to the sole structure **100**. The footwear **10** may further include an anterior end **12** associated with a forward-most point of the footwear, and a posterior end **14** corresponding to a rearward-most point of the footwear **10**. As shown in FIG. 3, a longitudinal axis A_{10} of the footwear **10** extends along a length of the footwear **10** from the anterior end **12** to the posterior end **14** parallel to a ground surface, and generally divides the footwear **10** into a medial side **16** and a lateral side **18**. Accordingly, the medial side **16** and the lateral side **18** respectively correspond with opposite sides of the footwear **10** and extend from the anterior end **12** to the posterior end **14**. As used herein, a longitudinal direction refers to the direction extending from the anterior end **12** to the posterior end **14**, while a lateral direction refers to the direction transverse to the longitudinal direction and extending from the medial side **16** to the lateral side **18**.

The article of footwear **10** may be divided into one or more regions. The regions may include a forefoot region **20**, a mid-foot region **22**, and a heel region **24**. The forefoot region **20** may be subdivided into a toe portion **20_T** corresponding with phalanges and a ball portion **20_B** associated with metatarsal bones of a foot. As shown, the article of footwear **10** may be described in terms of a metatarsophalangeal (MTP) axis A_{MTP} corresponding to an MTP joint of the foot, which generally extends between the toe portion **20_T** and the ball portion **20_B**. The mid-foot region **22** may correspond with an arch area of the foot, and the heel region **24** may correspond with rear portions of the foot, including a calcaneus bone.

With reference to FIGS. 2-4, the sole structure **100** includes a forefoot plate **102** attached to the upper **200** in the forefoot region **20** and a heel plate **104** attached to the upper **200** in the heel region **24**. The sole structure **100** and/or the plates **102**, **104** of the sole structure **100** may be described as including a top surface facing the upper and a bottom surface **108** formed on an opposite side of the sole structure **100** from the upper **200**. An outer peripheral edge **110** connects the top surface **106** to the bottom surface **108** and defines an outer peripheral profile of the sole structure **100**.

In the illustrated example, the forefoot plate **102** and the heel plate **104** are formed as separate components such that the forefoot plate **102** extends from a first end **112** at the anterior end **12** to a second end **114** adjacent to the mid-foot region **22**. Here, a portion of the peripheral edge **110** defining the second end **114** of the forefoot plate **102** extends

along a concave path from the medial side 16 to the lateral side 18 such that the second end 114 includes an arcuate recess 116 defining a pair of posterior-facing lobes 118 on opposite sides of the second end 112. The heel plate extends from a first end 120 adjacent to the mid-foot region 22 to a second end 122 at the posterior end 14. While the illustrated example includes the forefoot plate 102 and the heel plate 104 as separate components attached at opposite ends of the sole structure 100, the plates 102, 104 may be provided as a unitary component extending along an entire length of the article of footwear 10 from the anterior end 12 to the posterior end 14.

In the illustrated example, the sole structure 100 includes a plurality of directional traction elements 124-124f and an omnidirectional traction element 126. In this example, the directional traction elements 124-124f include unidirectional traction elements 124-124f provided as elongate members configured to move translationally through a ground surface (e.g., turf, soil) with a lower directional force in one direction than in an opposite direction, while the omnidirectional traction element 126 is provided as a round (i.e., cylindrical, conical, hemispherical) member configured to move through the ground surface in all directions with a substantially similar directional force.

With reference to FIGS. 2-4, each of the forefoot plate 102 and the heel plate 104 includes a plurality of the unidirectional traction elements 124-124f. As described herein, the unidirectional traction elements 124-124f may be alternatively referred to as blade cleats 124-124f. The blade cleats 124-124f may each be described as having a height extending from a proximal end 130 at the bottom surface 108 of the sole structure 100 to a distal end 132 spaced apart from the bottom surface 108 of the sole structure 100. Thus, the proximal end 130 of each blade cleat 124-124f forms a base 130 of the blade cleat 124-124f while the distal end 132 forms a tip 132 of the blade cleat 124-124f configured to engage the ground surface.

With continued reference to FIGS. 2-4, a length of each of the blade cleats 124-124f extends from a first end 134 to a second end 136 disposed at an opposite end of the blade cleat 124-124f from the first end 134. Each blade cleat 124-124f further includes a pair of side surfaces 138, 140 formed on opposite sides of the blade cleat 124-124f and extending from the first end 134 to the second end 136. Accordingly, a width of each blade cleat 124-124f is defined by a distance between the side surfaces 138, 140. As shown, a width of each of the blade cleats 124-124f tapers along a direction from the second end 136 to the first end 134. Here, the width of each blade cleat 124-124f tapers continuously along the entire length of the blade cleat 124-124f from the second end 136 to the first end 134. In other words, a width of the each blade cleat 124-124f is greater at the second end 136 than at the first end 134 such that the blade cleat 124-124f is configured to move through a ground surface material (e.g., soil) in a direction from the first end 134 to the second end 136 with a lower resistance than in a direction from the second end 136 to the first end 134.

In the illustrated example, each of the first surface 138 and the second surface 140 may be multi-faceted such that the blade cleats 124-124f each bend along a direction from the first end 134 to the second end 136. For instance, the first side surface 138 may include a first plurality of facets 142 arranged in series from the first end 134 to the second end 136. The facets 142 of the first side surface 138 are angled towards each other and cooperate to form a cupped or concave first surface 138, which may be referred to as an inner surface 138 of the blade cleat 124-124f. Conversely,

the second surface 140 may include a second plurality of facets 144 arranged in series from the first end 134 to the second end 136 on the opposite side of the blade cleat 124-124f from the first surface 138. The facets 144 of the second surface 140 are angled away from each other and cooperate to provide the second surface 140 with a convex shape. Thus, the second surface 140 may be referred to as an outer surface 140. In the illustrated example, the inner surface 138 and the outer surface 140 each include a pair of the facets 142, 144 such that each blade cleat 124-124f may be described as including first and second segments. However, in other examples, a facet resolution of the inner surface 138 and/or the outer surface 140 may be increased such that the surfaces 138, 140 include a greater number of facets 142, 144 or are fully arcuate.

As discussed in greater detail below, the inner and outer surfaces 138, 140 of the blade cleats 124-124f are configured to be aligned along one or more rotational radii of the sole structure 100 such that the surfaces are substantially aligned along one or more rotational paths associated with the pivot zone Z_P . Thus, the elongate shapes of the blade cleats 124-124f provided by the tapering width and the curved surfaces 138, 140 facilitate movement of the blade cleats 124-124f through the ground surface along the direction of the side surfaces 138, 140 with relatively low resistance while providing a high level of resistance (i.e., traction) in a direction transverse to the side surfaces 138, 140.

Optionally, each of the blade cleats 124-124f may include a chamfer 146 connecting the distal end 132 and the first end 134 of the blade cleat 124-124f. When included, the chamfer 146 includes a surface formed at an oblique angle between the distal end 132 and the first end 134 of the blade cleat 124-124f. The chamfer 146 provides the blade cleat 124-124f with a shorter length at the distal end 132 of the blade cleat 124-124f than at the base 130 of the blade cleat 124-124f such that the blade cleat 124-124f is configured to progressively engage the ground surface as the blade cleat 124-124f is inserted into the ground surface.

In some examples, the blade cleats 124-124f include caps 148 attached at the distal end 132 and, when present, the chamfer 146. Here, the caps 148 include a different material than the blade cleat 124-124f and are configured to tune an interface between the blade cleats 124-124f and the ground surface. For instance, the caps 148 may include materials having a lower durometer or a higher coefficient of friction than the body of the blade cleat 124-124f to provide the blade cleats 124-124f with better traction on relatively hard ground surfaces. Alternatively, the caps 148 may include materials having a higher durometer than a material of the blade cleats 124-124f to provide each of the blade cleats 124-124f with a hard tip for engaging softer ground surfaces.

With continued reference to FIGS. 2-4, the omnidirectional traction element 126 has a height extending from a proximal end 150 at the bottom surface 108 of the sole structure 100 to a distal end 152 spaced apart from the bottom surface 108 of the sole structure 100. Thus, the proximal end 150 forms a base 150 of the omnidirectional traction element 126 and the distal end 152 forms a tip 152 of the omnidirectional traction element.

Unlike the unidirectional traction elements 124-124f, which are substantially elongate in shape, the omnidirectional traction element 126 has a length and width that are substantially similar such that the omnidirectional traction element 126 is configured to move through the ground surface in all directions with substantially equal force or resistance. In the illustrated example, the omnidirectional traction element 126 is configured as a post cleat 126 having

a substantially flat distal end **152**. Specifically, the post cleat **126** is frustoconical such that a width or diameter of the post cleat **126** tapers along a direction from the base **150** to the tip **152**. In other examples, the post cleat **126** may be cylindrical, hemispherical, or have an equilateral polygonal cross section.

Referring to FIGS. 3 and 4, the forefoot plate **102** of the present example includes a first annular cleat group **154** disposed generally in the toe portion **20_T** of the forefoot region **20** and a second annular cleat group **156** arranged through the ball portion **20_B** of the forefoot region **20**. Here, the first annular cleat group **154** includes a plurality of the blade cleats **124a-124d** and a single one of the post cleats **126** all arranged in series within a first annular zone Z_{154} circumscribing the central pivot zone Z_P associated with the centroids C_{P1} , C_{P2} , C_{P3} discussed above. Similarly, the second annular cleat group **156** includes a pair of the blade cleats **124e**, **124f** arranged in series within a second annular zone Z_{156} that is concentric with the first annular zone Z_{154} . Here, the first annular zone Z_{154} is formed between a first minor radius R_{154a} and a first major radius R_{154b} and the second annular zone Z_{156} is formed between a second minor radius R_{156a} that is greater than the first major radius R_{154a} and a second major radius R_{156b} . Thus, while the traction elements **124a-124f**, **126** of each of the annular cleat groups **154**, **156** are not aligned along a common radius, the traction elements **124a-124f**, **126** of each cleat group **154**, **156** are aligned within a radius range defined between the minor radii R_{154a} , R_{156a} and the major radii R_{154b} , R_{156b} .

The first annular cleat group **154** may be referred to as an inner annular cleat group **154** and includes the post cleat **126** disposed immediately adjacent to the peripheral edge **110** on the medial side **16**. The post cleat **126** is disposed between the anterior end **12** and the MTP axis A_{MTP} . The inner annular cleat group **154** includes four of the blade cleats **124a-124d** arranged in series around the first annular zone Z_{154} . As shown, the blade cleats **124a-124d** are arranged in the same rotational direction around the first annular zone Z_{154} such that the first ends **134** of each one of the blade cleats **124b-124d** face the second ends **136** of a preceding one of the blade cleats **124a-124c** while the inner surface **138** of each blade cleat **124a-124d** faces inwardly towards the pivot zone Z_P . Thus, the blade cleats **124a-124d** are arranged to move in a rotational direction around the pivot zone Z_P .

Starting at the post cleat **126**, the blade cleats **124a-124d** are arranged in order including a first blade cleat **124a** on the medial side **16** of the anterior end **12**, a second blade cleat **124b** on the lateral side **18** of the anterior end **12**, a third blade cleat **124c** immediately adjacent to the peripheral edge **110** on the lateral side **18**, and a fourth blade cleat **124d** adjacent to the longitudinal axis A_{10} and the MTP axis A_{MTP} . As shown in FIG. 4, the first blade cleat **124a** is oriented such that the first end **134** faces the post cleat **126** on the medial side **16** and the outer surface **140** of the first blade cleat **124a** is adjacent and substantially parallel to the peripheral edge **110** of the forefoot plate **102**. The first end **134** of the second blade cleat **124b** faces the second end **136** of the first blade cleat **124a** and the outer surface **140** converges with the peripheral edge **110** along a direction from the first end **134** to the second end **136**. Thus, the first end **134** of the second blade cleat **124b** is spaced apart from the peripheral edge **110** by a greater distance at the first end **134** than at the second end **136**. The third blade cleat **124c** is disposed immediately adjacent to the peripheral edge **110** on the lateral side **18** and diverges from the peripheral edge **110** along the direction from the first end **134** to the second

end **136**. The fourth blade cleat **124d** is disposed in a central portion of the sole structure **100** adjacent to the longitudinal axis A_{10} and is oriented such that the first end **134** is closer to the anterior end **12** and the medial side **16** than the second end **136**.

With continued reference to FIGS. 3 and 4, the second annular cleat group **156** includes a pair of the blade cleats **124e**, **124f** arranged around the second annular zone Z_{156} . As provided above, the second annular zone Z_{156} is concentric with the first annular zone Z_{154} and has a larger radius than the first annular zone Z_{154} such that the blade cleats **124e**, **124f** of the second annular cleat group **156** partially surround the first annular cleat group **154**. Here, the second annular zone Z_{156} is sized such that the blade cleats **124e**, **124f** of the second annular cleat group **156** are disposed on a posterior side of the MTP axis A_{MTP} (i.e., in the ball portion **20_B**). The blade cleats **124e**, **124f** of the second annular cleat group **156** are arranged in the same rotational direction (e.g., clockwise, counterclockwise) around the pivot zone Z_P as the blade cleats **124a-124f** of the first annular cleat group **154**. For example, the first ends **134** of each of the blade cleats **124e**, **124f** of the second annular cleat group **156** face the lateral side **18** of the sole structure **100**. A first one of the blade cleats **124e** of the second annular cleat group **156** is disposed adjacent to the medial side **16** and a second one of the blade cleats **124f** of the second annular cleat group **156** is disposed adjacent to the lateral side **18**. The inner surfaces **138** of each of the blade cleats **124e**, **124f** are oriented towards the pivot zone Z_P .

With reference to FIGS. 5-5C, the positions of one of the inner blade cleats **124c**, one of the outer blade cleats **124e**, and the post cleat **126** are illustrated relative to the radii of rotation associated with the pivot centroids C_{P1} , C_{P2} , C_{P3} . As provided above, the first pivot centroid C_{P1} (FIG. 1A) is associated with a 45° outside cut, the second pivot centroid C_{P2} (FIG. 1B) is associated with a 180° outside cut (FIG. 1B), and the third pivot centroid C_{P3} (FIG. 1C) is associated with a 90° inside cut (e.g., accelerating, decelerating, or a planting kick).

With reference to FIG. 5A, the third blade cleat **124c** of the inner annular cleat group **154** is shown with the radii of rotation associated with the pivot centroids C_{P1} , C_{P2} , C_{P3} overlaid. Here, the length of the third blade cleat **124c** (i.e., from the first end **134** to the second end **136**) is substantially tangentially aligned along the radii of rotation associated with the first pivot centroid C_{P1} and the third pivot centroid C_{P3} and is oriented at an acute angle relative to the radius of rotation associated with the second pivot centroid C_{P2} . Thus, the third blade cleat **124c** is configured to move in the rotational directions corresponding to the first and third centroids C_{P1} , C_{P3} .

FIG. 5B shows the position of the post cleat **126** of the inner annular cleat group **154** with the radii of rotation of the pivot centroids C_{P1} , C_{P2} , C_{P3} overlaid. As shown, in Area 5B, the radii of rotation of the pivot centroids C_{P1} , C_{P2} , C_{P3} are substantially misaligned such that there is no best-fit orientation for a unidirectional blade cleat **124**. In other words, incorporating a directional traction element at Area 5B to accommodate rotational movement about one of the centroids C_{P1} , C_{P2} , C_{P3} would result in a relatively high degree of rotational resistance (i.e., traction) to rotational movement about the other two centroids C_{P1} , C_{P2} , C_{P3} . Accordingly, the omnidirectional post cleat **126** is placed at Area 5B to accommodate each of the different radii of rotation of the centroids C_{P1} , C_{P2} , C_{P3} .

In FIG. 5C, the medial side blade cleat **124e** of the outer annular cleat group **156** is shown with the radii of rotation of

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the pivot centroids C_{P1} , C_{P2} , C_{P3} overlaid. Here, the radii of rotation of the pivot centroids C_{P1} , C_{P2} , C_{P3} have a relatively high degree of tangential alignment such that the medial blade cleat **124e** is oriented to align the inner and outer surfaces **138**, **140** with each of the radii of rotation of the pivot centroids C_{P1} , C_{P2} , C_{P3} . Thus, the medial blade cleat **124e** is configured to move in the rotational directions corresponding to the first, second, and third centroids C_{P1} , C_{P2} , C_{P3} .

With continued reference to FIG. 5, the blade cleats **124e-124f** of the forefoot plate **102** are positioned at areas of the forefoot plate **102** where the radii of rotation of at least two of the pivot centroids C_{P1} , C_{P2} , C_{P3} are aligned. For instance, in addition to the blade cleats **124c**, **124e** discussed with respect to FIGS. 5A and 5C, FIG. 5 shows the first blade cleat **124a** of the inner annular cleat group **154** located where the radii of rotation for the first and third pivot centroids C_{P1} , C_{P3} are aligned while the second blade cleat **124b** of the inner annular cleat group **154** is located where the radii of rotation for the second and third pivot centroids C_{P1} , C_{P2} are aligned. The fourth blade cleat **124d** of the inner annular cleat group **154** and the lateral blade cleat **124f** of the outer annular cleat group **156** are each positioned at areas with relatively high degrees of alignment between the radii of rotation of all three of the pivot centroids C_{P1} , C_{P2} , C_{P3} .

As shown in FIG. 3, the heel plate **104** includes a plurality of the blade cleats **124** each oriented along the same rotational direction about the pivot zone Z_P as the blade cleats **124a-124f** of the forefoot plate **102**. In the illustrated example, the heel plate **104** includes two pairs of the blade cleats **124**, with a first pair aligned longitudinally along the lateral side **18** of the longitudinal axis A_{10} and a second pair aligned longitudinally along the medial side **16** of the longitudinal axis A_{10} . Here, all of the blade cleats **124** of the heel plate **104** include the first ends **134** facing towards the lateral side **18** of the sole structure **100**. The blade cleats **124** on the medial side **16** of the longitudinal axis A_{10} are oriented such that the second end **136** faces the medial side **16** and is closer to the anterior end **12** than the first end **134**. Similarly, the blade cleats **124** on the lateral side **18** of the longitudinal axis A_{10} are oriented such that the first end **134** faces the lateral side **18** and is closer to the anterior end **12** than the second end **136**.

As provided above, all of the blade cleats **124-124f** of the forefoot plate **102** and the heel plate **104** are oriented in the same rotational direction (i.e., clockwise, counterclockwise) such that the inner surfaces **138** and the outer surfaces **140** are substantially tangential to concentric radii of rotation about the pivot zone Z_P . Thus, the tapered, elongate, and bent shapes of the blade cleats **124** allow the sole structure **100** to rotate about the pivot zone Z_P with a minimized torsional force while the inner surfaces **138** and the outer surfaces **140** of the blade cleats **124** are configured to provide increased traction in lateral and longitudinal directions of the sole structure **100**.

With particular reference to FIGS. 6-8, an article of footwear **10a** is provided and includes a sole structure **100a** and the upper **200** attached to the sole structure **100a**. In view of the substantial similarity in structure and function of the components associated with the article of footwear **10** with respect to the article of footwear **10a**, like reference numerals are used hereinafter and in the drawings to identify like components while like reference numerals containing letter extensions are used to identify those components that have been modified.

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In the example of the sole structure **100a** shown in FIGS. 6-8, a forefoot plate **102a** includes an inner annular cleat group **154a** and an outer annular cleat group **156a** that are similar to the inner annular cleat group **154** and the outer annular cleat group **156** discussed previously. However, in this example, the radii R_{154a} , R_{154b} , R_{156a} , R_{156b} of the annular zones Z_{154} , Z_{156} have been reduced such that the inner and outer annular cleat groups **154a**, **156a** are more tightly grouped around the pivot zone Z_P . For example, the post cleat **126** and the first, second, and third blade cleats **124a-124c** are offset inwardly from the peripheral edge **110** of the sole structure **100**. Furthermore, the blade cleats **124e**, **124f** of the outer annular cleat group **156a** are each positioned on an anterior side of the MTP axis A_{MTP} (i.e., in the toe portion **20r**).

By reducing the respective radii of the cleat groups **154a**, **154b**, a torsional force required to rotate or twist the cleat groups **154a**, **154b** about the pivot zone Z_P through the ground surface is reduced relative to a torsional force associated with the sole structure **100** discussed previously. However, while the torsional forces associated with the sole structure **100a** are reduced, the rotational configuration of the traction elements **124**, **126** provides translational (e.g., lateral, longitudinal) traction forces comparable to the forces provided by the sole structure **100**, as the inner and outer surfaces **138**, **140** of the blade cleats **124** cooperate to engage the ground surface in different translational directions.

With particular reference to FIGS. 9-11, an article of footwear **10b** is provided and includes a sole structure **100b** and the upper **200** attached to the sole structure **100b**. In view of the substantial similarity in structure and function of the components associated with the article of footwear **10** with respect to the article of footwear **10b**, like reference numerals are used hereinafter and in the drawings to identify like components while like reference numerals containing letter extensions are used to identify those components that have been modified.

The sole structure **100b** of FIGS. 9-11 is substantially similar to the sole structure **100** discussed above and includes an inner annular cleat group **154b** and an outer annular cleat group **156b** arranged around annular zones Z_{154b} , Z_{156b} . As with the sole structure **100** discussed above, the inner annular cleat group **154b** is arranged along an annular zone Z_{154} such that the traction elements **124b-124d**, **160a-160b** of the inner annular cleat group **154b** are adjacent to the peripheral edge **110** of the forefoot plate **102b** and the traction elements **124f**, **160c** of the outer annular cleat group **156b** are positioned on a posterior side of the MTP axis A_{MTP} (i.e., in the ball portion **20b**).

The sole structure **100b** of the present example includes a plurality of bi-directional traction elements **160a-160c** in addition to the unidirectional traction elements **124b-124d**, **124f**. For example, where the sole structure **100** discussed above includes the post cleat **126**, the first blade cleat **124a** of inner annular cleat group **154**, and the medial blade cleat **124e** of the outer annular cleat group **156**, the sole structure **100b** of the present example includes bi-directional traction elements **160a-160c** incorporated into the annular cleat groups **154b**, **156b**. Generally, the bi-directional traction elements **160a-160c** are formed as symmetrical wing cleats **160a-160c** configured to allow bi-directional movement along a lengthwise direction of the traction element **160a-160c**. The bi-directional traction elements **160a-160c** and the unidirectional traction elements **124b-124d**, **124f** may be collectively referred to as directional traction elements.

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With reference to FIG. 9, each of the wing cleats **160a-160c** includes height extending from a proximal end **162** at the bottom surface **108** of the sole structure **100** to a distal end **164** spaced apart from the bottom surface **108** of the sole structure **100**. Thus, the proximal end **162** forms a base of the wing cleat **160a-160c** and the distal end **164** forms a tip of the wing cleat **160a-160c**.

With continued reference to FIGS. 9-11, a length of each of the wing cleats **160a-160c** extends from a first end **166** to a second end **168** disposed at an opposite end of the wing cleat **160a-160c** from the first end **166**. Each wing cleat **160a-160c** further includes a pair of side surfaces **170, 172** formed on opposite sides of the wing cleat **160a-160c** and extending from the first end **166** to the second end **168**. Accordingly, a width of each wing cleat **160a-160c** is defined by a distance between the side surfaces **170, 172**. As shown, the width of each of the wing cleats **160a-160c** tapers along the length from an intermediate portion towards each end **166, 168**. In other words, a width of the each wing cleat **160a-160c** is greater in a central portion than at the first end **166** and second end **168** such that the wing cleat **160** is configured to slice through a ground surface material (e.g., soil) in a direction towards either end **166, 168**.

In the illustrated example, each of the first side surface **170** and the second side surface **172** may be multi-faceted such that the wing cleats **160a-160c** each bend along a direction from the first end **166** to the second end **168**. For instance, the first side surface **170** may include a first plurality of facets **174** arranged in series from the first end **166** to the second end **168**. The facets **174** of the first side surface **170** are angled towards each other and cooperate to form a cupped or concave first side surface **170**, which may be referred to as an inner surface **170** of the wing cleat **160a-160c**. Conversely, the second surface **172** may include a second plurality of facets **176** arranged in series from the first end **166** to the second end **168** on the opposite side of the wing cleat **160a-160c** from the first side surface **170**. The facets **176** of the second side surface **172** are angled away from each other and cooperate to provide the second side surface **172** with a convex shape. Thus, the second side surface **172** may be referred to as an outer surface **172**. In the illustrated example, the inner side surface **170** and the outer side surface **172** each include three of the facets **174, 176** such that each wing cleat **160a-160c** may be described as including first and second end segments extending from an intermediate segment. However, in other examples, a facet resolution of the inner surface **170** and/or the outer surface **172** may be increased such that the surfaces **170, 172** include a greater number of facets **174, 176** or are fully arcuate.

Optionally, each of the wing cleats **160a-160c** may include a pair of chamfers **178** connecting the distal end **164** and with each of the first end **166** and the second end **168**. When included, the chamfers **178** include a surface formed at an oblique angle relative to the distal end **164** and each of the first end **166** and the second end **168** of the wing cleat **160a-160c**. The chamfers **178** provide the wing cleat **160a-160c** with a shorter length at the distal end **164** of the wing cleat **160a-160c** than at the base **162** of the wing cleat **160** such that the wing cleat **160** is configured to progressively engage the ground surface as the wing cleat **160a-160c** is inserted into the ground surface.

In some examples, the wing cleats **160a-160c** include caps **180** attached at the distal end **164** and, when present, the chamfers **178**. Here, the caps **180** include a different material than the body of the wing cleats **160a-160c** and are configured to tune an interface between the wing cleats **160a-160c** and the ground surface. For instance, the caps

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180 may include materials having a lower durometer or a higher coefficient of friction to provide the wing cleats **160a-160c** with better traction on relatively hard ground surfaces. Alternatively, the caps **180** may include materials having a higher durometer than a material of the wing cleats **160a-160c** to provide each of the wing cleats **160a-160c** with a hard tip for engaging softer ground surfaces.

The following Clauses provide an exemplary configuration of a sole structure and an article of footwear described above.

Clause 1: A sole structure for an article of footwear, the sole structure comprising: a first plurality of traction elements including a first series of directional traction elements arranged within a first annular zone in a first rotational direction around a pivot zone; and a second plurality of traction elements including a second series of directional traction elements arranged within a second annular zone concentric with and larger than the first annular zone, the second plurality of traction elements arranged in the first rotational direction around the pivot zone.

Clause 2: The sole structure of Clause 1, wherein each of the directional traction elements is elongate and includes a length extending from a first end to a second end along the first rotational direction.

Clause 3: The sole structure of Clause 2, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

Clause 4: The sole structure of Clause 2 or 3, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.

Clause 5: The sole structure of Clause 4, wherein the inner surface converges with the outer surface along the first rotational direction.

Clause 6: The sole structure of any one of Clauses 1-5, further comprising a third plurality of directional traction elements disposed in a heel region, each of the directional traction elements of the third plurality of directional traction elements oriented in the first rotational direction.

Clause 7: The sole structure of any one of Clauses 1-6, wherein the first plurality of traction elements and the second plurality of traction elements are disposed in a forefoot region of the sole structure.

Clause 8: The sole structure of any one of Clauses 1-7, wherein the first plurality of traction elements further includes an omnidirectional traction element arranged within the first annular zone.

Clause 9: The sole structure of Clause 8, wherein the omnidirectional traction element is disposed on a medial side of the sole structure and at least one of the directional traction elements of the first series is disposed on a lateral side of the sole structure.

Clause 10: The sole structure of any one of Clauses 1-9, wherein at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

Clause 11: A sole structure for an article of footwear, the sole structure comprising: a first annular group of traction elements arranged in series along a first annular zone in a forefoot region, the first annular group including a first directional traction element on a lateral side of the sole structure and a second directional traction element on a medial side of the sole structure; and a second annular group of traction elements arranged in series along a second annular zone concentric with the first annular zone, the second annular group of traction elements including a third

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directional traction element on the lateral side of the sole structure and a fourth directional traction element on the medial side of the sole structure.

Clause 12: The sole structure of Clause 11, wherein each of the directional traction elements is elongate and includes a length extending from a first end to a second end along a first rotational direction around a pivot zone of the sole structure.

Clause 13: The sole structure of Clause 12, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

Clause 14: The sole structure of Clause 12 or 13, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.

Clause 15: The sole structure of Clause 14, wherein the inner surface converges with the outer surface along the first rotational direction.

Clause 16: The sole structure of any one of Clauses 11-15, further comprising a third group of directional traction elements disposed in a heel region, each of the directional traction elements of the third group of directional traction elements oriented in the first rotational direction.

Clause 17: The sole structure of any one of Clauses 11-16, wherein the first annular group and the second annular group are disposed in a forefoot region of the sole structure.

Clause 18: The sole structure of any one of Clauses 11-17, wherein the first annular group of traction elements further includes an omnidirectional traction element arranged along the first annular zone.

Clause 19: The sole structure of Clause 18, wherein the omnidirectional traction element is disposed on a medial side of the sole structure.

Clause 20: The sole structure of any one of Clauses 11-19, wherein at least one of directional traction elements includes a unidirectional traction element and at least one of the directional traction elements includes a bidirectional traction element.

The foregoing description has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular configuration are generally not limited to that particular configuration, but, where applicable, are interchangeable and can be used in a selected configuration, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

What is claimed is:

1. A sole structure for an article of footwear, the sole structure comprising:

a first plurality of traction elements including a first series of directional traction elements including a first plurality of blade cleats arranged within a first annular zone and each of the first plurality of traction elements being oriented in a first rotational direction around a pivot zone, each blade cleat of the first plurality of blade cleats tapering continuously from a respective first end to a respective second end; and

a wing cleat aligned with traction elements of the first plurality of traction elements in the first rotational direction, the wing cleat tapering from an intermediate portion to a first end and from the intermediate portion to a second end located on an opposite side of the wing cleat than the first end.

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2. The sole structure of claim 1, wherein the wing cleat is disposed proximate to a medial side of the sole structure.

3. The sole structure of claim 1, wherein each of the blade cleats is elongate and includes a length extending from the respective first end to the respective second end along the first rotational direction.

4. The sole structure of claim 1, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the respective first end and the respective second end.

5. The sole structure of claim 1, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.

6. The sole structure of claim 5, wherein the inner surface converges with the outer surface along the first rotational direction.

7. The sole structure of claim 1, wherein the first plurality of traction elements and the wing cleat are disposed in a forefoot region of the sole structure.

8. The sole structure of claim 1, wherein the wing cleat is disposed between a first traction element of the first plurality of traction elements and a second traction element of the first plurality of traction elements along a length of a longitudinal axis of the sole structure.

9. The sole structure of claim 8, wherein a distance between the wing cleat and the first traction element is different than a distance between the wing cleat and the second traction element.

10. The sole structure of claim 1, wherein the wing cleat is symmetric.

11. A sole structure for an article of footwear, the sole structure comprising:

a first plurality of traction elements including a first series of directional traction elements including a first plurality of blade cleats arranged within a first annular zone and each of the first plurality of traction elements being oriented in a first rotational direction around a pivot zone; and

a wing cleat aligned with traction elements of the first plurality of traction elements in the first rotational direction and including a first facet, a second facet, and a third facet, the third facet disposed between the first facet and the second facet and cooperating with the first facet and the second facet to provide the wing cleat with a concave surface opposing at least one blade cleat of the first plurality of blade cleats.

12. The sole structure of claim 11, wherein the wing cleat is disposed proximate to a medial side of the sole structure.

13. The sole structure of claim 11, wherein each of the directional traction elements is elongate and includes a length extending from a first end to a second end along the first rotational direction.

14. The sole structure of claim 13, wherein each of the directional traction elements includes a chamfer formed adjacent to at least one of the first end and the second end.

15. The sole structure of claim 13, wherein each of the directional traction elements includes a concave inner surface and a convex outer surface each extending along the first rotational direction.

16. The sole structure of claim 15, wherein the inner surface converges with the outer surface along the first rotational direction.

17. The sole structure of claim 11, wherein the first plurality of traction elements and the wing cleat are disposed in a forefoot region of the sole structure.

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18. The sole structure of claim **11**, wherein the wing cleat is disposed between a first traction element of the first plurality of traction elements and a second traction element of the first plurality of traction elements along a length of a longitudinal axis of the sole structure.

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19. The sole structure of claim **18**, wherein a distance between the wing cleat and the first traction element is different than a distance between the wing cleat and the second traction element.

20. The sole structure of claim **11**, wherein the wing cleat is symmetric.

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